

SEMINAR SUMMARY:
APRIL 21, 2026 - BRENT CODY
CONNECTIONS BETWEEN GRAPH THEORY AND MUSIC

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We start with an introduction to modular synthesizers. But we are quickly confronted with a problem about maximal evenness, presenting a 12-cycle C_{12} , we are curious about the maximal placing of 5 onsets over the cycle of 12. We try to space the onsets alternatively at first, and we get a section of three uncovered vertices in the cycle. We do a little fiddling, and we end up with the repeating pattern of the black-keys on a piano over the white keys on the piano. The set $J_{12,5}^0 = \{0, 2, 4, 7, 9\}$ is the maximally even subset of C_{12} .

So naturally we ask, what does it mean to be maximally even? We come up with the following definition

Definition 1. (Clough, Douthett, 1991) Let $d(u, v)$ represent the distance from vertex u to vertex v of shortest length where $u, v \in C_n$, with C_n being the cycle on n -vertices. We call a set of vertices $\{a_0, a_1, \dots, a_{m-1}\} \subseteq C_n$ *maximally even* if whenever there is $1 \leq k \leq \lfloor m/2 \rfloor$, we have

$$d(a_i, a_{i+k}) \in \left\{ \left\lfloor \frac{nk}{m} \right\rfloor, \left\lceil \frac{nk}{m} \right\rceil \right\}$$

Now we notice something curious, namely that

$$J_{12,5} = \left\{ \left\lfloor \frac{12i}{5} \right\rfloor : 0 \leq i < 5 \right\}.$$

So this theorem arises:

Theorem 1. (Clough, Douthett, 1991) A set $A \subseteq C_n$ with $|A| = m$ is maximally even if and only if it is of the form

$$A = J_{n,m}^r = \left\{ \left\lfloor \frac{ni+r}{m} \right\rfloor : 0 \leq i < m \right\}.$$

And it seems that quite naturally the complement of a maximally even set is also maximally even

$$C_n \setminus J_{m,n}^r = J_{n,m-n}^{n-r-1}$$

Through the rest of the talk we talked about minimal energy from maximal energy, a curiosity indeed. I suppose a system run at full throttle can only maintain itself for so long

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before buckling. After buckling a system can continue to run, but needs help and fixing after buckling has occurred. Think of an engine perhaps you throw a rod, well there are 5 cylinders still running, but are you going to get the same performance. By no means. You have to stop, perform reparations on the engine, take parts out, put new parts in, clean the system. The same it seems is true of humans.